EXPERIMENT NO. 3

AIM: To create an Employee–Department–Project database schema, insert at least 30 employees across 5 departments and 8 projects, and perform SQL queries demonstrating **selection, projection, aggregate functions, GROUP BY, HAVING, CASE expressions, and ORDER BY**.

SOURCE CODE:

```
CREATE DATABASE companydb;
USE companydb;

CREATE TABLE Department(
   dept_id INT PRIMARY KEY,
   dept_name VARCHAR(30),
   location VARCHAR(30)
);

CREATE TABLE Employee(
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(40),
   salary DECIMAL(10,2),
   dept_id INT,
   manager_id INT NULL,
   FOREIGN KEY(dept_id) REFERENCES Department(dept_id)
);

CREATE TABLE Project(
   project_id INT PRIMARY KEY,
   project_name VARCHAR(40),
   dept_id INT,
   budget DECIMAL(12,2),
   FOREIGN KEY(dept_id) REFERENCES Department(dept_id)
);

CREATE TABLE Employee_Project(
   emp_id INT,
   project_id INT,
   PRIMARY KEY(emp_id,project_id),
   FOREIGN KEY(emp_id) REFERENCES Employee(emp_id),
   FOREIGN KEY(project_id) REFERENCES Project(project_id)
);

INSERT INTO Department VALUES
```

(1,'IT','Delhi'),(2,'HR','Lucknow'),(3,'Finance','Mumbai'),
(4,'Marketing','Pune'),(5,'AI','Bangalore');

INSERT INTO Employee VALUES
(1,'Aarav',55000,1,NULL),(2,'Ananya',65000,1,1),
(3,'Riya',48000,1,1),(4,'Karan',72000,1,1),
(5,'Neha',45000,1,2),(6,'Rahul',58000,1,2),
(7,'Ishita',62000,2,NULL),(8,'Aditya',52000,2,7),
(9,'Priya',47000,2,7),(10,'Rohan',68000,2,7),
(11,'Sneha',50000,2,8),(12,'Vivek',56000,2,8),
(13,'Meera',75000,3,NULL),(14,'Arjun',60000,3,13),
(15,'Nisha',54000,3,13),(16,'Varun',65000,3,13),
(17,'Kriti',49000,3,14),(18,'Yash',57000,3,14),
(19,'Tanya',70000,4,NULL),(20,'Dev',55000,4,19),
(21,'Pooja',46000,4,19),(22,'Manav',62000,4,19),
(23,'Simran',51000,4,20),(24,'Kabir',59000,4,20),
(25,'Aditi',85000,5,NULL),(26,'Nakul',70000,5,25),
(27,'Ira',68000,5,25),(28,'Aryan',78000,5,25),
(29,'Diya',60000,5,26),(30,'Mohit',65000,5,26);

INSERT INTO Project VALUES
(101,'Website',1,200000),(102,'CloudApp',1,300000),
(103,'Recruitment',2,150000),(104,'Payroll',3,250000),
(105,'Campaign',4,180000),(106,'Analytics',5,400000),
(107,'Chatbot',5,350000),(108,'FinanceAI',3,450000);

INSERT INTO Employee_Project VALUES
(1,101),(2,101),(3,102),(4,102),(5,101),
(7,103),(8,103),(9,103),(13,104),(14,104),
(15,108),(19,105),(20,105),(21,105),(25,106),
(26,106),(27,107),(28,107),(29,106),(30,107);

-- Selection
SELECT * FROM Employee WHERE salary > 60000;

emp_id	emp_name	salary	dept_id	manager_id
2	Ananya	65000.00	1	1
4	Karan	72000.00	1	1
7	Ishita	62000.00	2	null
10	Rohan	68000.00	2	7
13	Meera	75000.00	3	null
16	Varun	65000.00	3	13
19	Tanya	70000.00	4	null
22	Manav	62000.00	4	19
25	Aditi	85000.00	5	null
26	Nakul	70000.00	5	25
27	Ira	68000.00	5	25
28	Aryan	78000.00	5	25
30	Mohit	65000.00	5	26

-- Projection
SELECT emp_name, salary FROM Employee;

emp_name	salary
Aarav	55000.00
Ananya	65000.00
Riya	48000.00
Karan	72000.00
Neha	45000.00
Rahul	58000.00
Ishita	62000.00
Aditya	52000.00
Priya	47000.00
Rohan	68000.00
Sneha	50000.00
Vivek	56000.00
Meera	75000.00
Arjun	60000.00

-- Aggregate
SELECT COUNT(*) AS Total_Employees, AVG(salary) AS Average_Salary,
MAX(salary) AS Maximum_Salary, MIN(salary) AS Minimum_Salary
FROM Employee;

Total_Employees	Average_Salary	Maximum_Salary	Minimum_Salary
30	60233.333333	85000.00	45000.00

-- GROUP BY
SELECT dept_id, COUNT(*) AS Employees, AVG(salary) AS Avg_Salary
FROM Employee GROUP BY dept_id;

dept_id	Employees	Avg_Salary
1	6	57166.666667
2	6	55833.333333
3	6	60000.000000
4	6	57166.666667
5	6	71000.000000

-- HAVING
SELECT dept_id, AVG(salary) AS Avg_Salary
FROM Employee GROUP BY dept_id
HAVING AVG(salary) > 60000;

dept_id	Avg_Salary
5	71000.000000

-- CASE
SELECT emp_name, salary,
CASE
WHEN salary >= 70000 THEN 'High'
WHEN salary >= 55000 THEN 'Medium'
ELSE 'Low'
END AS Salary_Level
FROM Employee;

emp_name	salary	Salary_Level
Aarav	55000.00	Medium
Ananya	65000.00	Medium
Riya	48000.00	Low
Karan	72000.00	High
Neha	45000.00	Low
Rahul	58000.00	Medium
Ishita	62000.00	Medium
Aditya	52000.00	Low
Priya	47000.00	Low
Rohan	68000.00	Medium
Sneha	50000.00	Low
Vivek	56000.00	Medium
Meera	75000.00	High
Arjun	60000.00	Medium
Varun	65000.00	Medium
Kriti	49000.00	Low
Yash	57000.00	Medium
Tanya	70000.00	High
Dev	55000.00	Medium
Pooja	46000.00	Low
Manav	62000.00	Medium
Simran	51000.00	Low
Kabir	59000.00	Medium
Aditi	85000.00	High
Nakul	70000.00	High
Ira	68000.00	Medium
Aryan	78000.00	High
Diya	60000.00	Medium
Mohit	65000.00	Medium

-- ORDER BY
SELECT emp_name, salary FROM Employee
ORDER BY salary DESC;

emp_name	salary
Aditi	85000.00
Aryan	78000.00
Meera	75000.00
Karan	72000.00
Nakul	70000.00
Tanya	70000.00
Ira	68000.00
Rohan	68000.00
Varun	65000.00
Mohit	65000.00
Ananya	65000.00
Ishita	62000.00
Manav	62000.00
Diya	60000.00
Kabir	59000.00
Rahul	58000.00
Yash	57000.00
Vivek	56000.00
Dev	55000.00
Aarav	55000.00
Nisha	54000.00
Aditya	52000.00
Simran	51000.00
Sneha	50000.00
Kriti	49000.00
Riya	48000.00
Priya	47000.00
Pooja	46000.00
Neha	45000.00